



Solar power

Task A: Calculating the cost of electricity

- 1) A house with 4 or more bedrooms consumes an average of 11.8 kWh of electricity per day. In 2024, the average cost of electricity in the UK could be calculated as:

24.5 p per kWh consumed + 60 p daily standing charge

- a) How much electricity would this house consume per year?

$$11.8 \times 365 = 4307 \text{ kWh}$$

- b) What would be the daily cost of electricity for this house?

$$24.5 \times 11.8 + 60 = 349.1 \text{ p} = \text{£}3.49$$

- c) What would be the monthly cost of electricity for this house (based on 30.5 days)?

$$(24.5 \times 11.8 + 60) \times 30.5 = 10\,647.55 \text{ p} \approx \text{£}106.48$$

- d) What would be the annual cost of electricity for this house?

$$(24.5 \times 11.8 + 60) \times 365 = 127\,421.5 \text{ p} \approx \text{£}1274.22$$

Task B: Calculating possible savings with solar panels

- 1) A house consumes an average of 11.8 kWh of electricity per day.

The cost of obtaining electricity from a supplier company can be calculated using the following formula:

cost = 24.5 p per kWh consumed + 60 p daily standing charge

The owners install 8 solar panels, each with a power rating of 380 watts.

The panels get 4.5 hours of peak sunlight per day, on average.

The performance of the panels is reduced by 25% due to factors such as shade.

How much money do the occupants save each year on their electricity bills by having solar panels?

$$E = PH$$

E = electricity produced (kWh)

P = power rating of panel (kW)

H = amount of peak sunlight (h)

Cost if all electricity was provided by their supplier:

$$(24.5 \times 11.8 + 60) \times 365 = 127\,421.5 \text{ p}$$

Electricity produced by 8 solar panels per day:

$$0.38 \times 4.5 \times 0.75 \times 8 = 10.26 \text{ kWh}$$

Remaining electricity required:

$$11.8 - 10.26 = 1.54 \text{ kWh}$$

Cost of electricity with solar panels:

$$(24.5 \times 1.54 + 60) \times 365 = 35\,671.45 \text{ p}$$

Money saved per year:

$$127421.5 - 35671.45 = 91\,750.05 \text{ p} \approx \text{£}917.50$$



Task C: Calculating the cost of installing solar panels

- 1) The occupants of a house buy 8 solar panels.

The costs involved with buying and installing the panels are shown below.

Price per panel: £200

Storage battery: £4000

Other required electrical equipment: £1700

Labour costs: £1500

- a) How much will it cost in total to buy and install the solar panels? £8800
- b) Their annual electricity bill without solar panels was £1275.
Their annual electricity bill with solar panels is £350.
How much money do they save per year by having solar panels? £925
- c) How many years will it take for their annual savings to sum to more than the upfront cost of buying and installing the solar panels? $8800 \div 925 = 9.513...$ (10 years)

- 2) The occupants of a house are considering investing in solar panels.

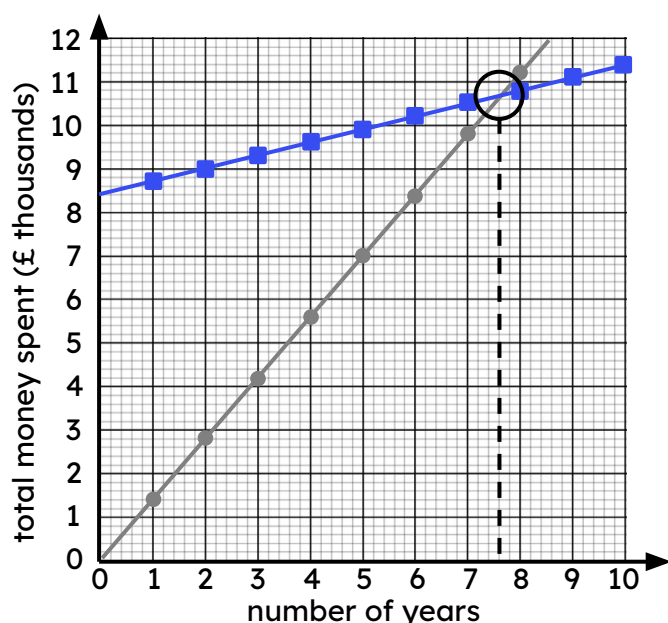
Previously, they obtained all their electricity from their energy supplier company, with an annual bill of £1400.

The solar panels cost £8400 to buy and install.

Once the panels are installed, their annual electricity bill is estimated to be £300.

- a) Complete the table below to show the total money spent after each year.

number of years	0	1	2	3	4	5	6	7	8	9	10
total cost without solar (£ thousands)	0	1.4	2.8	4.2	5.6	7	8.4	9.8	11.2	12.6	14
total cost with solar (£ thousands)	8.4	8.7	9	9.3	9.6	9.9	10.2	10.5	10.8	11.1	11.4



- b) Use your answer from part (a) to plot a graph showing total money spent after each year **without** solar panels.

—●— without solar panels

- c) Use your answer from part (a) to plot a graph showing total money spent after each year **with** solar panels.

—■— with solar panels

- d) How many years will it take for the solar panels to pay for themselves?

7.6 years or 8 years

